The documentation and process conversion measures necessary to comply with this document shall be completed by 6 February 2025.

INCH-POUND

MIL-PRF-19500/246L
w/AMENDMENT 2
6 September 2024
SUPERSEDING
MIL-PRF-19500/246L
w/AMENDMENT 1
31 May 2024

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, TYPES 1N3289, 1N3291, 1N3293, 1N3294, 1N3295, AND R TYPES, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon power rectifiers. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Normal and reverse types (reverse types, suffix R). Reverse and normal types are identical except the normal types have the cathode connected to the stud and the reverse types have the anode connected to the stud. Designated values are applicable to both types.

1.3 Package outlines. The device package for the encapsulated device type are as follows: (DO-205AA - formerly DO-8) in accordance with figure 1.

* 1.4 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	V_{RSM}	V_{RWM}	I_O T_C = 150°C	I_O T_C = 134°C (1)	I_{FSM} 1/120 s	Barometric pressure (reduced)	T_J and T_{STG}
	V (pk)	V (pk)	A dc	A dc	A (pk)	mm Hg	°C
1N3289, R	240	200	75	100	1,600		-65 to +200
1N3291, R	480	400	75	100	1,600	8	-65 to +200
1N3293, R	720	600	75	100	1,600	16	-65 to +200
1N3294, R	960	800	75	100	1,600	30	-65 to +200
1N3295, R	1,200	1,000	75	100	1,600	54	-65 to +200

(1) Derate linearly 1.5 A dc/°C for $T_C \geq 134$°C.

1.4.1 Thermal resistance characteristic: $R_{\theta JC}$ = 0.4°C/W, see figure 2.

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1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, and "JANTXV".

1.5.2 Device type. The designation system for the device types of semiconductors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the semiconductors covered by this specification sheet are as follows: "3289", "3291", “3293”, “3294” and "3295".

1.5.3 Suffix symbols. The following suffix letters are incorporated in the PIN in the order listed in the table as applicable:

R	An R suffix symbol indicates the reverse type that has the anode connected to the stud (see figure 1).

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

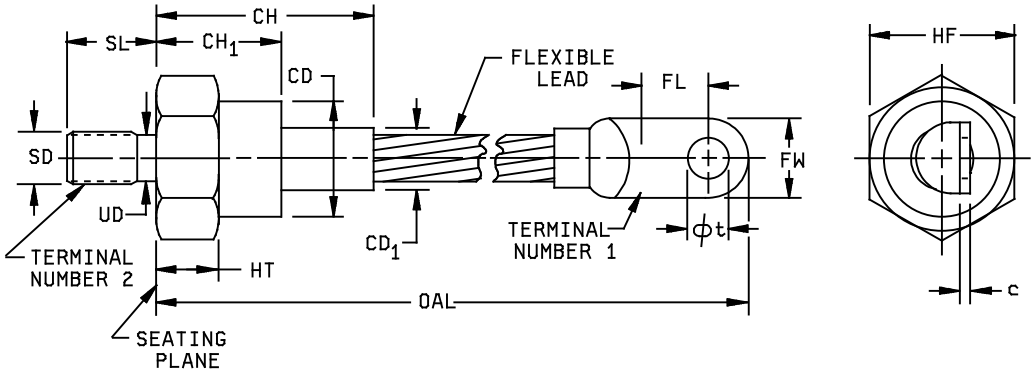
DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.625	1.000	15.88	25.40	8
CD_1		.500		12.70	
CH		1.750		44.45	
CH_1		1.140		28.96	
c	.050	.120	1.27	3.05	
FL	.300	.450	7.62	11.43	6
FW		.670		17.02	
HF	1.031	1.063	26.19	27.00	
HT	.125	.500	3.18	12.70	5
OAL	4.300	5.065	109.22	128.65	
SD					4
SL	.605	.645	15.37	16.38	
UD	.343	.373	8.71	9.47	
ϕt	.250	.310	6.35	7.87	4

NOTES:
1. Dimensions are in inches.
2. Millimeter equivalents are given for general information only.
3. Complete threads to extend to within 2.5 threads of seating plane.
4. .375-24 UNF-2A. Maximum pitch diameter of plated threads shall be basic pitch diameter (.3479 inch (8.837 mm) reference.
5. A chamfer or undercut on one or both ends of hexagonal portions is optional.
6. Minimum flat.
7. For marking (see 3.5).
8. The body of the device, with the exception of the hexagon and flexible lead extensions, lies within cylinder defined by CD_1 and CH, CD_1 not to exceed actual HF.
9. Terminal shape is optional.

* FIGURE 1. Physical dimensions.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (DO-205AA) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, and as specified herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Devices shall be marked as specified in MIL-PRF-19500.

3.6 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end or the use of other techniques considered commercial practice.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E qualification. Group E qualification shall be performed herein for qualification or re-qualification only. In case qualification was awarded to a prior revision of the associated specification that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot to this revision to maintain qualification.

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4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with appendix E, table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screening	Measurements for JANTX and JANTXV levels
1	Method 2073 of MIL-STD-750 may be used in lieu of method 2074 for compression bonded devices only.
3c	Thermal impedance (see 4.3.1) and figure 2.
4	5,000 G; performed prior to installation of external lead.
10	Not applicable.
11	I_{RM1} and V_{FM} of subgroup 2 of table I herein.
* 12	Method 1038, condition A; $V_{RM} = V_{RWM}$ rated (see 1.4), T_C = 150°C; f = 50 - 60 Hz.
13	Subgroup 2 of table I herein: ΔI_{RM1} = ±0.5 mA (dc) or 100 percent of initial value, whichever is greater. ΔV_{FM} = ±0.1 V (dc).

4.3.1 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD}, t_{SW}, (and V_H where appropriate). See table II, group E, subgroup 4 and figure 2 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table E-V of MIL-PRF-19500, table I herein, and as specified herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 herein.

4.4.2.1 Group B inspection, appendix E, table E-VIB (JAN, JANTX, and JANTXV of MIL-PRF-19500).

	Subgroup	Method	Condition
*	B3	1048	Blocking life. T_C = +150°C, -10, +0; $V_{RM} = V_{RWM}$ (see 1.4); t = 340 hours. f = 50 – 60 Hz.
	B3	1037	See 4.5.1, ΔT = 85°C minimum; I_O = ≥ 75 A, 5,000 cycles. (Separate samples may be used.)
	B5	3101 or 4081	$R_{\theta JC}$ = 0.4°C/W.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500.

4.4.3.1 Group C inspection, appendix E, table E-VII of MIL-PRF-19500.

	Subgroup	Method	Condition
	C2	1051	-55°C to +175°C, 25 cycles.
	C2	2036	Tension (lead): Test condition A; weight = 100 pounds; t = 15 ±3 seconds. Torque (terminal): Test condition D1; 10 inch-pounds; t = 15 ±3 seconds. Torque (stud): Test condition D2; 125 inch-pounds; t = 15 ±3 seconds.
	C2	1071	Gross leak. test condition D.
*	C6	1048	Blocking life. T_C = +150°C, -10, +0; V_{RM} = V_{RWM} (see 1.4), t = 1,000 hours f = 50 - 60 Hz.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and table II herein. Delta limits in accordance with table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Power cycling. One complete test shall be as follows: First, heat the case to the maximum temperature specified by passing the specified amount of forward current through the diode under test. Then, remove the applied current and allow the case temperature to cool to the minimum case temperature specified. No time limit is applicable to any one cycle, but the cycling shall be continuous until the required number of cycles have been completed. Sample size = 12, c = 0.

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TABLE I. Group A inspection. 1/

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Forward voltage	4011	Condition B, Pulse method: I_{FM} = 310 A (pk); pulse width = 8.3 ms max; duty cycle = 2 percent max	V_{FM1}		1.55	V (pk)
Reverse current	4016	DC method or equivalent pulse: Condition A or B.	I_{RM1}			
1N3289, R		V_{RM} = 200 V			10	mA
1N3291, R		V_{RM} = 400 V			10	mA
1N3293, R		V_{RM} = 600 V			10	mA
1N3294, R		V_{RM} = 800 V			10	mA
1N3295, R		V_{RM} = 1,000 V			10	mA
Reverse current at peak reverse voltage	4016	DC method or equivalent pulse: Condition A or B.	I_{RM2}			
1N3289, R		V_{RM} = 240 V			40	mA
1N3291, R		V_{RM} = 480 V			40	mA
1N3293, R		V_{RM} = 720 V			40	mA
1N3294, R		V_{RM} = 960 V			40	mA
1N3295, R		V_{RM} = 1,200 V			40	mA
Thermal impedance	3101	See 4.3.1 and figure 2	$Z_{\theta JX}$			°C/W
Subgroup 3						
High temperature operation:		T_C = +200°C				
Reverse current	4016	DC method or equivalent pulse: Condition A or B.	I_{RM3}			
1N3289, R		V_{RM} = 200 V			30	mA
1N3291, R		V_{RM} = 400 V			30	mA
1N3293, R		V_{RM} = 600 V			30	mA
1N3294, R		V_{RM} = 800 V			30	mA
1N3295, R		V_{RM} = 1,000 V			30	mA

See footnote at end of table.

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TABLE I. Group A inspection - Continued. 1/

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Low temperature operation:		T_C = -65°C				
Forward voltage	4011	Condition B, Pulse method: I_{FM} = 310 A (pk); pulse width = 8.3 ms max; duty cycle = 2 percent maximum	V_{FM2}		1.60	V (pk)
Reverse current:	4016	DC method or equivalent pulse: Condition A or B.	I_{RM4}			
1N3289, R		V_{RM} = 200 V			40	mA
1N3291, R		V_{RM} = 400 V			40	mA
1N3293, R		V_{RM} = 600 V			40	mA
1N3294, R		V_{RM} = 800 V			40	mA
1N3295, R		V_{RM} = 1,000 V			40	mA
Subgroups 4 and 5						
Not applicable						
Subgroup 6						
Surge current	4066	Condition A, T_C = +150°C; I_{FSM} = 1,600 A (pk) I_O = 75 A dc; ten 1/120 s surges; 1 surge/minute V_{RM} = V_{RWM} (see 1.4)				
Electrical measurements		Group A subgroup 2				
Subgroup 7						
Not applicable						

1/ For sample plan, see MIL-PRF-19500.

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TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Thermal shock	1056	100 cycles, 0°C to 100°C.	
Hermetic seal	1071		
Electrical measurement		See table I, group A, subgroup 2 and table III herein.	
Subgroup 2			45 devices c = 0
Steady-state reverse bias	1038	Test condition A, 1,000 hours.	
Electrical measurement		See table I, subgroup 2 and table III herein.	
Subgroup 4			Sample size = N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			3 devices c = 0
Barometric pressure	1001	V_{RWM} = rated V_{RWM} (see 1.4), pressure = rated pressure (see 1.4), t = 1 minute.	

TABLE III. Groups E delta measurements. 1/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage (change)	4011	Pulse method: I_{FM} = 310 A (pk) pulse width = 8.3 ms max; duty cycle = 2 percent maximum Condition B	ΔV_{FM}		+0.1	V (pk)

1/ The electrical measurements for appendix E, table E-IX of MIL-PRF-19500 are as follows:
a. Subgroup 1, see table III herein, step 1.
b. Subgroup 2, see table III herein, step 1.

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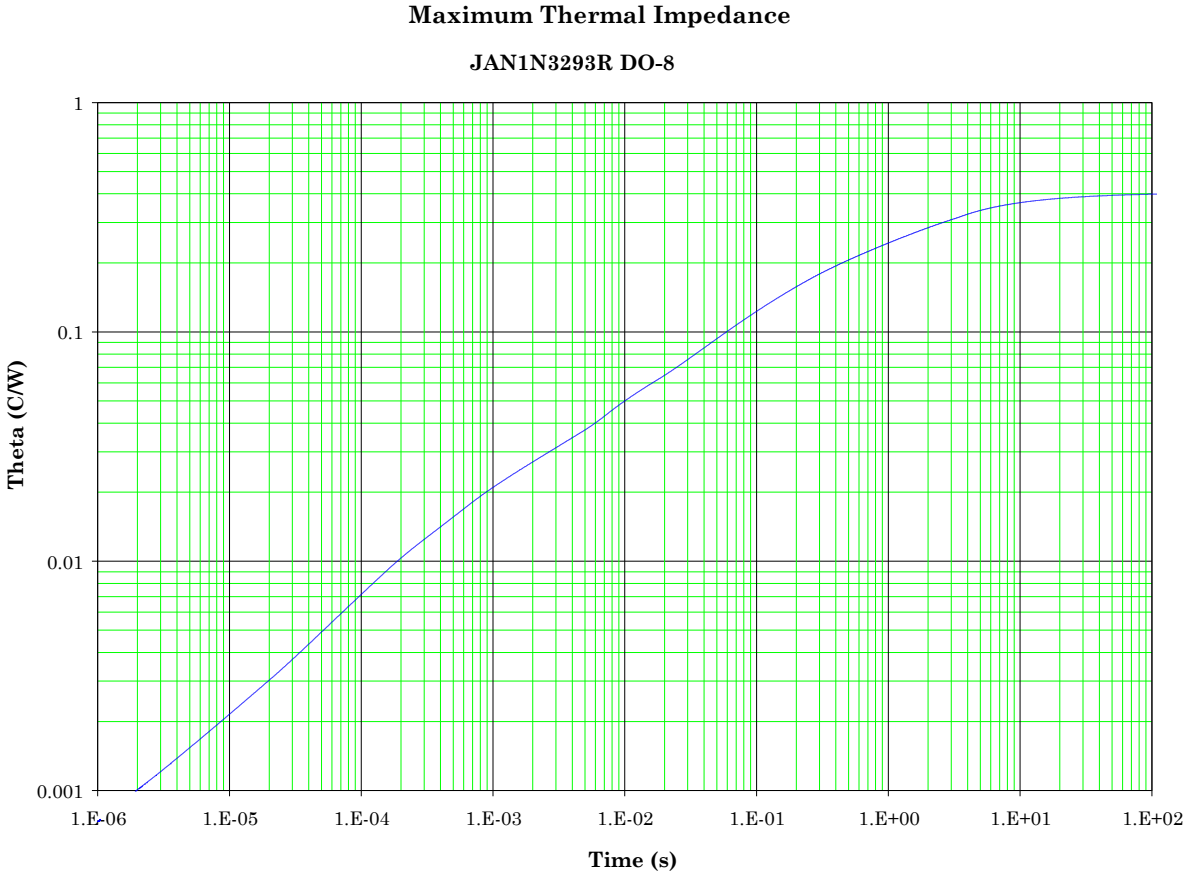


FIGURE 2. Thermal impedance graph ($R_{\theta JC}$) for DO205AA = .4°C/W, this curve takes into account both standard and reverse polarity.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

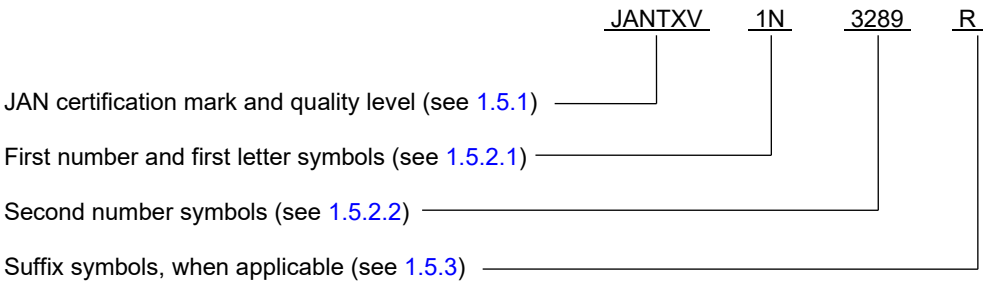
c. Lead finish (see 3.4.1).

d. The complete Part or Identifying Number (PIN), see title and section 1.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



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* 6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

PINs for types 1N3289, 1N3291, 1N3292, 1N3293, 1N3294, AND 1N3295.			
JAN1N3289 JAN1N3291 JAN1N3293 JAN1N3294 JAN1N3295	JANTX1N3289 JANTX1N3291 JANTX1N3293 JANTX1N3294 JANTX1N3295	JANTXV1N3289 JANTXV1N3291 JANTXV1N3293 JANTXV1N3294 JANTXV1N3295	
JAN1N3289R JAN1N3291R JAN1N3293R JAN1N3294R JAN1N3295R	JANTX1N3289R JANTX1N3291R JANTX1N3293R JANTX1N3294R JANTX1N3295R	JANTXV1N3289R JANTXV1N3291R JANTXV1N3293R JANTXV1N3294R JANTXV1N3295R	

6.6 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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